

Finding time complexity of algorithms

Problem 1: Finding Complexity using Counter Method:

Convert the following algorithm into a program and find its time complexity using the counter method.

void function (int n)

```
{  
    int i= 1;  
  
    int s =1;  
  
    while(s <= n)  
    {  
        i++;  
        s += i;  
    }  
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int count=0;
3  void fun(int n){
4      int i=1;
5      int s=1;
6      count+=2;
7      while(s<=n){
8          i++;
9          count++;
10         s+=i;
11         count++;
12         count++;
13     }
14     count++;
15     printf("%d",count);
16 }
17 int main(){
18     int n;
19     scanf("%d",&n);
20     fun(n);
21 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Problem 2: Finding Complexity using Counter method:

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int count=0;
3 void func(int n){
4     if(n==1){
5         count++;
6     }
7     else{
8         count++;
9         for(int i=1;i<=n;i++){
10             count++;
11             for(int j=1;j<=n;j++){
12                 count++;
13                 count++;
14                 count++;
15                 break;
16             }
17             count++;
18         }
19         count++;
20     }
21 }
22 int main(){
23     int n;
24     scanf("%d",&n);
25     func(n);
26     printf("%d",count);
27 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Problem 3: Finding Complexity using Counter Method:

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2 int count=0;
3 void factor(int n){
4     count++;
5     for(int i=1;i<=n;++i){
6         count++;
7         if(n%i==0){
8             count++;
9         }
10        count++;
11    }
12 }
13 int main(){
14     int n;
15     scanf("%d",&n);
16     factor(n);
17     printf("%d",count);
18 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Problem 4: Finding Complexity using Counter Method:

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int count=0;
3  void func(int n)
4  {
5      int c= 0;
6      count++;
7      count++;
8      for(int i=n/2; i<n; i++){
9          count++;
10         for(int j=1; j<n; j = 2 * j){
11             count++;
12             for(int k=1; k<n; k = k * 2){
13                 count++;
14                 c++;
15                 count++;
16             }
17             count++;
18         }
19         count++;
20     }
21 }
22 int main(){
23     int n;
24     scanf("%d",&n);
25     func(n);
26     printf("%d",count);
27 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Problem 5: Finding Complexity using counter method:

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2 int count=0;
3 void reverse(int n)
4 {
5     int rev = 0, remainder;
6     count++;
7     while (n != 0)
8     {
9         remainder = n % 10;
10        count++;
11        rev = rev * 10 + remainder;
12        count++;
13        n/= 10;
14        count++;
15        count++;
16    }
17    count++;
18    count++;
19 }
20 int main(){
21     int n;
22     scanf("%d",&n);
23     reverse(n);
24     printf("%d",count);
25 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.